# PunkBackup — iPhone Setup, Manual Build (Alternate Method)

> This manual shows the PC app in **English** — go to **"Settings"** → **"English"** if it isn't already switched (instant, no restart).

**This is the alternate path.** Almost everyone should use the main **iPhone Setup Manual** instead — it installs a ready-made Shortcut file in about a minute. Use this guide only if you'd rather build every action by hand, or want to understand exactly what each step does.

## Before you start, have these ready

Same requirement as the main manual: this system supports several devices/people (profiles) on the same PC, and **each device needs its own profile and its own token**.

On your PC: 1. Open the app and click **"Start backup"**. 2. In **"Settings"**, click **"+ Add profile"** and name it after THIS device (e.g. `iPhone de María`), then choose its destination folder when prompted. 3. The app shows you (and auto-copies) that profile's **token** — it only works for this one device.

Note down these values:

| Value | Where to find it in the app | Example |
|---|---|---|
| Server address (same for every profile) | "Address" field | `http:■/■/■[YOUR_PC_NAME].■local:■878` |
| Fallback IP (in case the address above doesn't resolve from the iPhone) | "Fallback IP" field | `http:■/■/■192.■168.■1.■50:■8787` |
| This profile's token | "Copy token" button on the profile row you just created | a long string of letters/numbers |

> Your iPhone and your PC must be on the **same WiFi network**.

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## Step 1 — Create the main Shortcut ("PunkBackup")

1. Open the **Shortcuts** app.
2. **My Shortcuts** tab → **+** button → **Add Action** (new shortcut).
3. Tap the name at the top and rename it to: `PunkBackup`.
4. Add the following actions **in this order** (search each by name using the magnifying glass):

### 1) Store the server address and token as variables

- **Text** action → type the server address, e.g. `http:■/■/■[YOUR_PC_NAME].■local:■8787`
- **Set Variable** action → name it `ServerURL` → value: the Text above.
- **Text** action → paste the Token **for this profile/device** you copied from the app.
- **Set Variable** action → name it `Token` → value: the Text above.

> These are the only two actions you'll ever need to edit again — for example if you regenerate this profile's token from the app ("Regenerate token" button).

### 2) Tell the server a backup run is starting

- **Get Contents of URL** action:
- URL: `ServerURL` + `/run/start` (tap the `ServerURL` variable, then type `/run/start` right after it).
- Method: **POST**
- Headers: add one → Key `X-Backup-Token` → Value: the `Token` variable.
- **Get Dictionary Value** action → Key: `run_id` → (applied to the result of the previous step).
- **Set Variable** action → name it `RunID` → value: the result above.

### 3) Set up the backward block-sweep

`Find Photos` has no built-in pagination — asked for everything at once, it fails outright on a large library (confirmed: no `Limit` = the Shortcut errors out, even doing nothing but counting results). The fix is to sweep the library **backward in bounded blocks**, newest photos first, each block capped at 50 items, advancing a moving date boundary between blocks — all within one run of the Shortcut.

- **Text** action → type `50` (how many blocks of 50 to sweep per run — see the tuning note at the end of this section).
- **Set Variable** → name it `Repeticiones`.
- **Current Date** action.
- **Adjust Date** action → **Add 1 day** to it (so the very first block excludes nothing — "before tomorrow" covers everything you have today).
- **Set Variable** → name it `Limite` (type: Date — keep it as an actual Date value here, not text; the next step compares it directly against each photo's `Date Taken`).
- **Repeat** action (the plain "Repeat X times" kind, **not** "Repeat with Each" — this is the outer loop) → for its count, insert the **Repeticiones** variable instead of typing a fixed number.

Everything below, through the end of section 4, goes **inside** this outer `Repeat`:

- **Find Photos** action (may be labeled "Filter Photos" on some iOS versions):
- Add a filter: **Date Taken** → **is before** → insert the **Limite** variable.

- Sort by: **Date Taken**. Order: **Latest First** (this is the opposite of what you'd naturally pick — it's what makes the sweep start from your newest photos and work backward).
- **Limit**: turn it **on**, set to **50**. This is mandatory, not optional — `Find Photos` without a `Limit` fails on a large library.
- **Count** action → **Items** in the `Find Photos` result above.
- **If** action: condition = the `Count` result **is** `0` (this fires once the sweep has passed your oldest photo — nothing left to check).
- Inside the "If": **Stop This Shortcut** action.
- Nothing needed in "Otherwise" — if the count isn't 0, execution just continues past the "End If" into section 4 below.

### 4) Loop through each item in the current block: ask first, only upload if missing

**Repeat with Each** action over the `Find Photos` result from section 3. Inside this inner "Repeat" block:

**Step a.0) Set Variable** action, as the very first action in this block:

- Value: tap "Repeat Item" and choose the **Date Taken** attribute (leave it as the raw Date — don't run it through Format Date here).
- Name the variable `UltimaFecha`. This captures the current item's exact date+time; by the time the loop finishes, it holds the OLDEST item's date in this block (since sorted newest-first) — section 4's closing steps use it to move `Limite` for the next block.

**Step a) Format Date** action:

- Date: tap "Repeat Item" and choose the **Date Taken** attribute.
- Format: **Custom** → type: `yyyy-MM-dd'T'HH:mm:ss`
- Store this as a variable named `TakenAt` (**Set Variable** action).
- **Fragile chip, double-check it after saving**: if you later add or move any action INSIDE this same "Repeat" block (e.g. step a.0 above), Shortcuts can silently reconfigure this chip to point at a different attribute (like "Name") instead of "Date Taken" — with no red error, no visible warning. The symptom shows up weeks later: ALL new photos land in the same folder (this month's) instead of their real month. If you suspect this, tap the chip inside "Format Date" and confirm it says **Date Taken**, not Name or anything else.

**Step a.2) Text** action — build the full filename, **with extension** (important: Shortcuts' "File Name" attribute on its own does **not** include the extension — skip this step and you'll end up with files like `IMG_1234` instead of `IMG_1234.HEIC`):

- In the text field, insert: **Repeat Item** → choose the **File Name** attribute → type a period `.` (no spaces) → insert **Repeat Item** again → choose the **File Extension** attribute.
- It should read something like: `[File Name].[File Extension]`
- Store this as a variable named `FileName` (**Set Variable** action).

**Step b) Get Contents of URL** action — the lightweight check, WITHOUT the file:

- URL: `ServerURL` + `/check`
- Method: **POST**
- Request Body: **Form**
- Form fields:
  - `filename` → value: the **FileName** variable (from step a.2 — NOT the bare "File Name" attribute, which is missing the extension).
  - `taken_at` → value: `TakenAt` variable.
- No need to send the file size — Shortcuts has no reliable way to give a plain byte count (it always formats it as something like "1.2 MB"), so the check only uses filename + date. The exact content check happens later, in `/upload`.
- Headers: `X-Backup-Token` → `Token` variable.
- **Get Dictionary Value** action → key `missing` → applied to that result.

**Step c) If** action: condition = the value above **has any value** (this is how the server tells you "it's missing, upload it"; when a file is already backed up, the server simply omits the `missing` field entirely, so this condition is automatically false and the "If" is skipped — you do NOT need "is equal to" or to type `false` anywhere, "has any value" is already the right option):

**Step c.0) If** (first action inside step c's "If" — decide whether this item needs Encode Media BEFORE attempting any upload, not as a retry after one fails):

Condition: tap **Repeat Item** → attribute **Media Type** → `is` → `Video`. (If your iOS version doesn't offer "Media Type" in the Repeat Item attribute list, use instead: **Repeat Item** → **File Extension** `is` `MOV`, then tap **+** to add more conditions joined by **Or**: `is` `MP4`, `Or` → `is` `M4V`.)

**Video branch** (inside this "If"):

- **Encode Media** action → Item: insert **Repeat Item** — once, and don't tap the chip again afterward. (Same care as always: tapping it a second time opens a "Name/Album/Width..." property menu instead of the raw item.)
- **Size**: `Passthrough` (doesn't reduce quality or resolution — it just forces Shortcuts to read the full file, which is exactly what fails for some videos otherwise — see the note below).
- **Set Variable** → name it `ItemFile` → value: the result of Encode Media above.

**Otherwise branch** (photo, not video):

- **Set Variable** → name it `ItemFile` → value: insert **Repeat Item** directly — once, and don't tap it again afterward.

**Step c.1) Get Contents of URL** (inside step c's "If", after step c.0 — this is now the only upload attempt):

- URL: tap the field and insert, IN THIS ORDER, inside the same text field: the **ServerURL** chip → type `/upload?filename=` → insert the **FileName** variable chip (the same one from step a.2 — NOT the bare "File Name" attribute, which is missing the extension) → type `&taken_at=` → the **TakenAt** chip → type `&run_id=` → the **RunID** chip. Inserting the chips directly into the URL

field (instead of building the text separately with "Combine Text") makes Shortcuts URL-encode them automatically.
- Method: **POST**
- **Request Body** → change it to **File** (no longer Form).
- In the value field that appears, insert the **`ItemFile`** variable (from step c.0) — NOT Repeat Item, NOT Encode Media directly. If this chip ever shows up highlighted in red/"broken" in the editor — it can happen after editing actions earlier in the same loop — delete it and re-insert it fresh; a broken reference here silently uploads an empty (0-byte) file instead of erroring visibly.
- Headers: `X-Backup-Token` → `Token` variable.

(Nothing needed in the outer "Otherwise" branch of step c — if it was already backed up, just move on to the next item.)

> **Why Encode Media runs proactively instead of as a retry**: confirmed on a real device, twice, two different failure shapes. First (2026-09-16): a video imported from WhatsApp/Messages/other apps sometimes arrives at the server as 0 bytes — Shortcuts fails to read its real data straight from the Photos library. Second, and the reason for this redesign (2026-10-23): some videos make the DIRECT upload attempt's own "Get Contents of URL" action fail outright, in the Shortcuts editor itself, with the alert `WFPhotoMediaContentItem produced no file representation for type public.mpeg-4` — a client-side failure to coerce the Photos item into file bytes at all, happening BEFORE any HTTP request is sent. The original design (try the raw item first, retry with Encode Media only if the server's response contained a `detail` field) could never catch this second failure shape, because there was no server response to inspect — the upload action itself never ran. Running `Encode Media` with `Size: Passthrough` unconditionally for every video (gated only by step b's `/check`, so an already-backed-up video never pays this cost) avoids both failure shapes at once, since it forces the full read/remux to happen before Shortcuts ever tries to attach the item as a request body. Confirmed harmless to quality (`ffprobe` comparison: same codec, resolution, bitrate — no real loss, see PLAN.md §5.1) and confirmed in production (2026-10-23): a live run uploaded videos up to 69.8 MB with zero 0-byte rejections and zero coercion errors. See AGENTS.md §5, lesson 28, for the full analysis, including why this doesn't create a data-loss risk even though it drops the same-run retry (the server never records a failed/partial upload as backed up, so `/check` simply catches it again on the next run).

Still inside the **inner** "Repeat with Each", after its own "End Repeat" but **before** the outer "End Repeat" from section 3 — these two actions advance the sweep to the next block:

- **Adjust Date** action → **Subtract 1 minute** from **UltimaFecha** (a small safety buffer, so a photo sharing the exact same timestamp as the block boundary — e.g. burst-mode shots — never gets silently skipped; worst case it's re-checked redundantly, which is harmless).
- **Set Variable** action → variable: **Limite** (pick the *existing* `Limite` from the list, don't create a new one with the same name) → value: the result of the `Adjust Date` above.

> **Tuning `Repeticiones`**: confirmed working with `Repeticiones` up to 50 (≈2,500 photos checked in one run) in real testing; `Repeticiones = 180` (≈9,000 photos) has also been confirmed not to crash or fail the Shortcut itself. Don't assume a block that's already fully backed up is fast just because it transfers

> no file — measured directly from real activity logs, an "already backed up" check can still take a real double-digit number of seconds per item, not a near-instant round trip. See the Troubleshooting Manual's "How fast is a backup, really?" section for the measured numbers (including a per-`Repeticiones` time table) and why a run that's mostly re-checking already-backed-up content can take longer than one uploading the same number of brand-new items.

> **Important, non-obvious limitation**: `Limite` always resets to "tomorrow" at the start of every run — there is no memory of where a *previous* run stopped. If a run ends because it used up all its `Repeticiones` (rather than because it found an empty block and reached your oldest photo), **running it again with the same `Repeticiones` makes no further progress** — it re-sweeps the exact same newest photos and stops at the exact same point every time. To actually reach older photos, you must **increase `Repeticiones`**, not just re-run the Shortcut. For a one-time full backfill of a large library, set `Repeticiones` high enough to cover your whole library in a single run (roughly `your total photo count ÷ 50`, rounded up — e.g. ~180 for a 9,000-photo library) rather than a small number you plan to re-run — whether iOS can sustain that many blocks in one run hasn't been confirmed for very large values, so increase it gradually and watch the PC app's activity log to see it's keeping up.

### 5) Close out the run and show you the result

- **Get Contents of URL** action:
- URL: `ServerURL` + `/run/finish`
- Method: **POST**, Form with field `run_id` = `RunID` variable.
- Header `X-Backup-Token` → `Token` variable.
- **Get Dictionary Value** actions (one per field: `files_new`, `files_skipped`, `files_conflict`) applied to that result.
- **Text** action to build a message, e.g.: `Backup complete \nNew: [files_new]\nAlready had: [files_skipped]\nConflicts: [files_conflict]`
- **Show Notification** (or **Show Result**) action with that text.

Save the shortcut — you can now tap it manually to test it. (See the "Tuning `Repeticiones`" note in section 3 for how long a run takes on a large library, and how to size the sweep for your first backfill.)

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## Step 2 — Short "check status" shortcut

Optional but recommended — lets you check status without running a full backup.

1. Create a new shortcut named `Backup Status`.
2. Repeat the two "Text" + "Set Variable" actions from Step 1.1 (`ServerURL`, `Token`).
3. **Get Contents of URL** action: URL = `ServerURL` + `/status`, Method **GET**, header `X-Backup-Token` → `Token`.
4. **Get Dictionary Value** actions for `last_backup_at` and `total_files_backed_up`.
5. **Show Notification** action with those values.

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## Step 3 — Automate it

Same as the main manual's automation step — once your hand-built `PunkBackup` shortcut works, go to the **iPhone Setup Manual**'s "Automate it: run on its own when you get home" section and follow it exactly the same way, regardless of how the Shortcut was built.

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## Notes

- The system **never deletes or modifies anything on your iPhone** — it only reads photos and uploads them.
- The truth about "what's already backed up" lives **only in the PC's destination folder**, never on the iPhone. If you switch your profile's folder/USB drive on the PC, the shortcut automatically re-sends whatever is missing from the new folder — you don't need to do anything different on the iPhone.
- Exact action names can vary slightly between iOS versions; if you can't find an action with the exact name above, search by a keyword instead (e.g. "dictionary", "URL", "repeat").